Guardeo Dashboard

Project Name : Guardio Dashboard
Client Name: Indian Army
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1. Project Overview

The **Guardeo Control Dashboard** is a modern, responsive fully offline control system designed to operate and monitor a robotic ground unit (robot dog) equipped with a camera system, pan-tilt gun mechanism, and environmental sensing modules.

The system provides a **secure, real-time control interface** that works seamlessly on **tablets, and touchscreen devices**. It enables operators to remotely control robot movement, monitor live video, manage weapon safety mechanisms, and perform calibration of critical hardware components.

2. Objectives

The main objectives of the system are:

- Provide a **real-time remote control dashboard** for robot operations
- Ensure **operator safety** through multi-layer action validation
- Deliver **responsive UI** for field usage on tablets and touch devices
- Enable **camera and weapon mechanism calibration** from the dashboard
- Display **live system status**, and robot status
- Support **secure login and session-based access control**
- **Deliver fully** Secure Gun trigger functionality using secure password.
- Provide as per range meter view for shoot.
- Joysticks for controlling robot.

3. System Architecture

Frontend (Client Side)

Built using:

- **HTML5** – Structure of dashboard
- **CSS3** – Responsive layout, grid system, touch-friendly UI
- **JavaScript** – Joystick logic, live controls, real-time UI updates

Backend (Server Side)

Built using:

- **Python 3**
- **Django Framework**
- Websocket APIs for robot communication
- Session-based authentication system

Robot Communication Layer

- WebSocket / API-based communication with robot SDK.

- Sends:
    - Movement commands
    - Pan-tilt controls
    - Calibration values
    - Safety commands

- Receives:
    - System status
    - Sensor alerts (e.g., pothole detection)
    - Camera stream status

4. Core Features

🎮 4.1 Dual Joystick Robot Control

The dashboard includes two virtual joysticks:

Joystick	Function
Left Joystick	Robot movement (forward, backward, forward-left,forward-right,backword-left,backword-right)
Right Joystick	Rotation control

Key Capabilities

- Fully responsive on **touchscreens devices and tablets.**
- Circular motion boundary for controlled movement
- Auto-centers when released
- Outputs **normalized motion values** ready for ROS/WebSocket integration
- Smooth control with no sudden jumps

🛡️ 4.2 Safety-Enabled Weapon Control

Weapon operation is designed with **strict safety logic**.

Safety Mechanisms

- Fire control is **password protected**
- Buttons remain **disabled by default**
- Visual indicators for:
  - Enabled state
  - Disabled state
  - Safety lock active
- Touch interaction disabled when safety is off
- Automatic reset after firing or safety condition trigger

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📹 4.3 Live Robot Camera View

The dashboard includes an embedded **live video panel**.

Features

- Real-time video streaming
- Auto-resizes based on screen size
- Centered layout using Flexbox/Grid
- Works in **portrait and landscape modes**
- Optimized for tablets used in field operations

⚙️ 4.4 Gun Mechanism Pan-Tilt Control

Operators can precisely control the weapon mechanism:

- **Pan (Left/Right)** control
- **Tilt (Up/Down)** control
- Smooth step-based movement
- Integrated **calibration mode**
    - Set center position
    - Adjust min/max angle limits
    - Save calibration values to backend

📹 4.5 Camera Pan-Tilt & Calibration

The camera module includes:

- Independent digital pan-tilt control
- Fine step adjustments
- Camera center alignment tool
- Calibration interface to:
    - Define movement limits
    - Reset to default
    - Save calibration values according to range eg.100 and 50 mtr.

🕳️ 4.6 Pothole Detection System

The robot includes a **sensor-based pothole detection module**.

Dashboard Integration

- Real-time pothole alert display
- Visual warning indicators
- Event logs sent from robot sensors
- Helps operator navigate terrain safely

🔋 4.7 Robot Status Dashboard

A real-time monitoring section displays:

- Battery percentage
- Robot active / inactive state
- System health status
- Communication status
- Sensor status

🧩 4.8 Security & Access Control

The system includes a built-in **security layer**:

- Login screen with **username/password**
- Session-based authentication (Django sessions)
- Unauthorized users cannot access dashboard routes
- Fire command protected with **secondary password**
- System auto-locks on logout or session expiry

🖥️ 4.9 System Controls

Operators can also:

- Shut down robot system remotely (with confirmation)
- Restart control modules
- View connection diagnostics

5.Technologies Used

Sr No	Layer	Technologies
1.	Frontend	HTML,CSS,JS
2.	Backend	Django-5,Python3.11
3.	Robot communication	ROS2
4.	Hardware & OS	Raspberry pi4 & Ubuntu 22

6. Installation & Setup Guide

a.cd project/myproj

b.pip install -r requirements.txt

c.python manage.py migrate

d.python3 manage.py runserver 0.0.0.0:8000

e.For auto start app we can setup service file in /etc/systemd/system

7.Project Directory

```
/myproj
|-App/
|  ├── __init__.py
|  ├── views.py # views functions for webpages
|  └── urls.py # urls endpoint
| ├── templates/
| |    ├── login.html
| |    ├── dashboard.html # Main dashboard interface
| ├── static/
| |  ├── css/
| | |   ├── login.css # Complete CSS for layout, responsiveness
| | |   ├── dashboard.css
| |  ├── js/
| |   ├── login.js # login page javascript
| |    ├── home.js # controls of joysticks
|  ├── images/
|      ├── icon-192.png # App icon
|      ├── guardeo-logo.png
|    ├── Indian_Army.png
├── myproj/
|  ├── settings.py # Django settings file
|  ├── urls.py # URL configuration for the entire project
├── manage.py # Command-line utility for administrative tasks
├── requirements.txt # List of dependencies required by the project
└── README.md
```

8. How to Operate the System

1. Open app from tablet Guardeo Dashboard or open http://army.local
2. Login with authorized credentials
    a. Username - admin
    b. Password - 9999
3. Verify robot connection status(Green - Active,Red-Inactive)
4. Monitor live camera feed
5. Use joysticks for Robot movement
6. Up,Down & Left,Right buttons for Gun mechanism Pan tilt
7. Enable weapon system using password (1234)
8. Perform calibration when required In calibration screen from drop down button.camera calibration for camera calibration
9. Monitor robot and system status

9. Conclusion

The **Guardeo Control Dashboard** delivers a **secure, responsive, and field-ready control interface** for robotic operations. It integrates **real-time control, monitoring, calibration, and safety enforcement** into a single platform built with modern web technologies and a robust Django backend.